Unified Smart Home Architecture

Project: G+1 Villa Unified Smart Home Solution

Scope: First floor, common areas, exterior, garden, and expandable ground floor

Document Type: Architecture + schematic design brief

1. Objective

Create one unified smart home platform for a Ground + 1 Floor villa that controls lighting, fans, AC, curtains, TVs, washrooms, exterior lights, garden lighting, and common-area automation from one system.

The system must support:

- Local control inside the house
- Remote control from outside the house
- Physical wall switch fallback
- Mobile app control
- Voice assistant integration
- Scene control
- Scheduling
- Future expansion for security, CCTV, water tank, irrigation, and energy monitoring

2. Property Automation Scope

2.1 Main Areas

Zone	Area	Automation Scope
First Floor	5 bedrooms	Strip lights, COB lights, fan, AC, TV, curtains, intercom provision
First Floor	5 attached washrooms	Strip lights, COB lights, exhaust fan
First Floor	Living area	Big TV, lighting, curtains, sunroof, AC/fan if required
First Floor	Dining area	Main lights, mood lights, optional curtains
First Floor	Prayer area	Soft scene lighting
First Floor	Powder room	Light and exhaust automation
Common	Staircase	Step lights, COB lights, strip lights, motion/night scene
Exterior	Front, rear, left, right	Facade, wall, gate, pathway, security lighting
Outdoor	Garden	Landscape lights, pathway lights, irrigation-ready control

2.2 Standard Bedroom Device List

Each bedroom should include:

- False ceiling strip light control
- COB light control by group
- Fan ON/OFF and speed control
- AC control through IR blaster or brand gateway
- LCD TV power/source control if required
- Motorized curtain control for each window/glass section
- Intercom provision
- Wall keypad/manual switch fallback
- Room scenes

2.3 Standard Washroom Device List

Each attached washroom should include:

- Strip light control
- COB light control
- Exhaust fan control
- Optional occupancy sensor
- Optional humidity sensor
- Auto exhaust timer

3. Recommended System Architecture

Use a hybrid local-first architecture. The house should continue working even when the internet is unavailable.

```
                        REMOTE ACCESS / CLOUD LAYER
+--------------------------------------------------------------------+
| Mobile App / Web Dashboard / Voice Cloud                           |
| User login, remote control, notifications, backup access           |
+--------------------------------+-----------------------------------+
                                 | HTTPS / MQTTS
                                 v
+--------------------------------------------------------------------+
| Cloud Broker + API                                                 |
| Authentication, remote relay, encrypted device messaging           |
+--------------------------------+-----------------------------------+
                                 | Secure outbound bridge
                                 v
                        LOCAL HOME EDGE LAYER
+--------------------------------------------------------------------+
| Smart Home Edge Gateway                                            |
| Mini PC / industrial controller / Home Assistant class gateway     |
| - Local automation engine                                          |
| - Local MQTT broker                                                |
| - Device registry                                                  |
| - Scene engine                                                     |
| - Local dashboard/API                                              |
+-------+----------------------------+------------------------+--------+
        |                            |                        |
        | Wi-Fi / Ethernet           | Zigbee / Thread        | IR / RS485 / dry contact
        v                            v                        v
+---------------+           +-----------------+       +-----------------+
| Room Modules  |           | Sensor Network  |       | AC / TV / AV    |
| Lights, fan,  |           | motion, temp,   |       | curtain, gate,  |
| curtains      |           | humidity        |       | intercom link   |
+---------------+           +-----------------+       +-----------------+
```

4. Network Architecture

4.1 Physical Network

```
ISP Fiber Router
      |
      +-- Main Network Switch
      |       +-- Smart Home Edge Gateway
      |       +-- Wi-Fi Access Points
      |       +-- CCTV NVR future provision
      |       +-- Admin/maintenance port
      |
      +-- Guest Wi-Fi isolated from smart home devices

Smart Home VLAN / Network
      +-- Room controllers
      +-- Curtain controllers
      +-- AC IR gateways
      +-- Exterior lighting controllers
      +-- Garden controllers
      +-- Sensors
```

4.2 Recommended Network Separation

Network	Purpose	Access
Main LAN	Owner phones, tablets, laptops	Full local access
Smart Home LAN/VLAN	Controllers, sensors, gateway	Restricted
Guest Wi-Fi	Visitors	Internet only
CCTV VLAN future	Cameras/NVR	Restricted

5. Protocol Architecture

Function	Recommended Protocol
Local control commands	MQTT over LAN
Remote control	HTTPS + secure MQTT bridge
Low-power sensors	Zigbee / Thread
High-load reliable controllers	Wired relay/dimmer modules
AC control	IR blaster or brand-specific API/gateway
Curtain motors	Relay/dry contact, RS485, or motor controller
TV control	IR, HDMI-CEC, or IP control where supported
Intercom integration	Dry contact/API only if supported by intercom brand

6. Complete System Block Diagram

```
+--------------------------------------------------------------------+
|                           USER CONTROL                             |
+---------------------+----------------------+-----------------------+
| Mobile App          | Wall Keypads         | Voice Assistant       |
| Local + Remote      | Manual fallback      | Optional              |
+----------+----------+----------+-----------+-----------+-----------+
           |                     |                       |
           v                     v                       v
+--------------------------------------------------------------------+
|                       LOCAL EDGE GATEWAY                           |
| Automation engine | MQTT broker | Scene engine | Device registry   |
+-----+---------------+------------------+-------------------+------+
      |               |                  |                   |
      v               v                  v                   v
+-----------+   +------------+    +--------------+    +---------------+
| Lighting  |   | Climate    |    | Curtains     |    | Sensors       |
| modules   |   | AC + fan   |    | motors       |    | motion/humid. |
+----+------+   +-----+------+    +------+-------+    +------+--------+
     |                |                  |                   |
     v                v                  v                   v
 COB/strip        AC, fan speed      Open/close/stop     Automation input
 lights           comfort scenes     position scenes     alerts and logic
```

7. Room Controller Schematic

One smart controller per bedroom is recommended. High-load circuits must be sized by a licensed electrician.

```
                      BEDROOM SMART CONTROLLER

  230V AC IN
     |
     +-- MCB / protection from DB
     |
     +-- Power supply 230V AC to 5V/12V DC
     |
     v
+----------------------------------------------------------------+
| Controller MCU / Smart Relay Module                            |
|                                                                |
| Inputs                                                         |
| - Manual switch/keypad input                                   |
| - Curtain limit/feedback input where available                 |
| - Optional motion/temp sensor                                  |
|                                                                |
| Outputs                                                        |
| - Relay 1: COB light group 1                                   |
| - Relay 2: COB light group 2                                   |
| - Dimmer/PWM: false ceiling strip light                        |
| - Fan relay/speed controller                                   |
| - Curtain open relay/dry contact                               |
| - Curtain close relay/dry contact                              |
| - IR output: AC / TV                                           |
|                                                                |
| Network                                                        |
| - Wi-Fi/Ethernet/Zigbee depending on selected module          |
+----------------------------------------------------------------+
     |             |             |             |
     v             v             v             v
 COB lights    Strip driver   Fan control   Curtain motor
```

8. Bedroom Wiring Concept

```
 DB Lighting Circuit
       |
       v
 Bedroom Switch Box / Automation Panel
       |
       +-- Smart relay channel 1 -- COB lights group A
       +-- Smart relay channel 2 -- COB lights group B
       +-- LED dimmer channel  ---- 12V/24V strip light driver
       +-- Fan controller      ---- Ceiling fan
       +-- Curtain channel A   ---- Curtain motor open
       +-- Curtain channel B   ---- Curtain motor close
       +-- IR blaster          ---- AC + TV

 Manual wall switches remain connected to controller inputs.
 If automation fails, manual control must still operate critical lights.
```

9. Washroom Controller Schematic

```
                    WASHROOM SMART CONTROLLER

  Lighting circuit from DB
          |
          v
+-------------------------------------------------+
| Washroom relay module                           |
|                                                 |
| Inputs                                          |
| - Manual switch                                 |
| - Optional occupancy sensor                     |
| - Optional humidity sensor                      |
|                                                 |
| Outputs                                         |
| - Relay 1: COB lights                           |
| - Relay 2: strip lights                         |
| - Relay 3: exhaust fan                          |
+-------------------------------------------------+
          |                  |                  |
          v                  v                  v
       COB lights        Strip lights      Exhaust fan
```

10. Living Area Schematic

```
                         LIVING AREA CONTROL

+------------------------------------------------------------------+
| Living area controller                                           |
|                                                                  |
| Lighting                                                         |
| - Main COB groups                                                |
| - Cove/strip lights                                              |
| - Feature wall lights                                            |
| - Staircase-linked lighting scene                                |
|                                                                  |
| Media                                                            |
| - Big LCD TV IR/IP control                                       |
| - Optional AV receiver                                           |
|                                                                  |
| Openings                                                         |
| - Curtain motor groups for glass areas                           |
| - Sunroof motor/open-close control if compatible                  |
|                                                                  |
| Scenes                                                           |
| - Guest, Movie, Relax, Bright, All Off                           |
+------------------------------------------------------------------+
```

11. Exterior And Garden Schematic

```
                         EXTERIOR CONTROL PANEL

             From dedicated exterior lighting MCB
                           |
                           v
+------------------------------------------------------------------+
| Outdoor-rated automation panel                                   |
|                                                                  |
| Outputs                                                          |
| - Front facade lights                                            |
| - Rear lights                                                    |
| - Left side wall lights                                          |
| - Right side wall lights                                         |
| - Gate/entry lights                                              |
| - Garden lights                                                  |
| - Pathway lights                                                 |
| - Irrigation pump/valve future provision                         |
|                                                                  |
| Inputs                                                           |
| - Motion sensors optional                                        |
| - Light sensor optional                                          |
| - Rain/soil sensor future provision                              |
+------------------------------------------------------------------+
```

12. Suggested Controller Distribution

Location	Controller Type	Notes
Bedroom 1	Room controller	Lights, fan, AC, TV, curtains
Bedroom 2	Room controller	Lights, fan, AC, TV, curtains
Bedroom 3	Room controller	Lights, fan, AC, TV, curtains
Bedroom 4	Room controller	Lights, fan, AC, TV, curtains
Bedroom 5	Room controller	Lights, fan, AC, TV, curtains
Each attached washroom	3-channel relay module	COB, strip, exhaust
Living area	Multi-channel controller	Lighting, curtains, TV, sunroof
Dining/prayer area	Lighting controller	Scene control
Staircase	Lighting + motion controller	Night automation
Powder room	3-channel relay module	Light and exhaust
Exterior panel	Outdoor-rated relay panel	Zone-wise exterior control
Garden	Outdoor controller	Lights and irrigation-ready

13. Room Naming And Device Naming Standard

Use consistent names from day one.

```
floor/area/device/function

Examples:
first/bedroom1/light/cob
first/bedroom1/light/strip
first/bedroom1/fan/main
first/bedroom1/ac/main
first/bedroom1/curtain/window1
first/washroom1/exhaust/main
exterior/front/light/facade
garden/light/pathway
```

14. MQTT Topic Structure

```
home/{floor}/{area}/{device}/{function}/set
home/{floor}/{area}/{device}/{function}/state
home/{floor}/{area}/{device}/{function}/telemetry
home/{floor}/{area}/{scene}/trigger
```

Examples:

```
home/first/bedroom1/light/cob/set
home/first/bedroom1/light/cob/state
home/first/bedroom1/ac/main/set
home/first/living/scene/movie/trigger
home/exterior/front/light/facade/set
```

15. Control Flow - Local Mode

```
User taps mobile app inside house
        |
        v
App detects local gateway
        |
        v
Command sent directly to local gateway
        |
        v
Local MQTT broker publishes command
        |
        v
Room controller receives command
        |
        v
Relay/dimmer/IR/curtain output executes
        |
        v
Device state returns to app
```

16. Control Flow - Remote Mode

```
User taps mobile app outside house
        |
        v
Command sent to cloud API
        |
        v
Cloud authenticates user
        |
        v
Cloud broker forwards command through secure bridge
        |
        v
Local gateway receives command
        |
        v
Local MQTT broker publishes command
        |
        v
Room controller executes
        |
        v
State sync returns to cloud and app
```

17. Scene Architecture

17.1 Bedroom Scenes

Scene	Action
Relax	Strip light warm/dim, COB low, fan medium, curtains partial close
Sleep	Lights off, fan set, AC sleep mode, curtains close
Wake Up	Curtains open, soft strip light, AC/fan adjust
Movie	COB off, strip dim, curtains close, TV on
All Off	Lights off, fan/AC optional, curtains unchanged or close

17.2 Common Scenes

Scene	Action
Good Morning	Selected curtains open, soft lights, exterior off
Good Night	Interior off, bedroom sleep scenes, exterior security on
Guest Mode	Living/dining/prayer lighting ready
Prayer Mode	Prayer area warm light, distractions off
Dining Mode	Dining lights warm and focused
Movie Mode	Living curtains close, lights dim, TV on
Away Mode	All non-essential loads off, exterior/security active
Vacation Mode	Randomized evening lights, exterior schedule

18. Security Architecture

```
+--------------------------------------------------------------+
| Security controls                                            |
+--------------------------------------------------------------+
| - Separate smart home network/VLAN                           |
| - No direct public port forwarding to local gateway          |
| - Remote access through secure cloud tunnel only             |
| - Strong admin password and multi-user roles                 |
| - Device certificates or unique tokens                       |
| - Encrypted cloud communication                              |
| - Local firewall on edge gateway                             |
| - Guest Wi-Fi isolated from automation devices               |
+--------------------------------------------------------------+
```

19. Safety Requirements

- All 230V AC work must be handled by a licensed electrician.

- Use certified relay/dimmer modules with correct load rating.

- Use separate MCBs for lighting, power, AC, exterior, and garden circuits.

- Use surge protection at DB level.

- Keep low-voltage control wiring separate from high-voltage wiring.

- Use outdoor-rated IP65/IP67 enclosures for exterior and garden controllers.

- Do not place non-waterproof modules inside washroom wet zones.

- Manual override must be available for essential lighting.

- AC loads must not be switched through small lighting relays.

20. Implementation Phases

Phase 1 - Survey

- Confirm complete floor plan
- Count COB lights per room
- Measure strip light lengths
- Count curtain motors
- Confirm AC brand/model per room
- Confirm fan type
- Confirm switchboard and DB locations
- Confirm exterior and garden lighting zones

Phase 2 - Electrical Design

- Circuit load calculation
- Relay/dimmer channel mapping
- DB/MCB mapping
- Switchboard module placement
- Cable routing
- Low-voltage and high-voltage separation

Phase 3 - Network Design

- Gateway location
- Router and switch location
- Wi-Fi access point placement
- Smart home VLAN setup
- Zigbee/Thread coordinator location if sensors are used

Phase 4 - Installation

- Install smart modules
- Install curtain motors
- Install sensors
- Install gateway
- Pair devices
- Configure rooms and scenes

Phase 5 - Testing

- Test manual switches
- Test app control

- Test internet-offline local control
- Test remote control
- Test scenes
- Test schedules
- Test safety fallback

21. Minimum BOQ Categories

Category	Items
Central control	Edge gateway, router/switch, backup power
Lighting control	Relay modules, dimmers, LED strip drivers
Fan control	Fan controllers or compatible smart regulators
AC control	IR blasters or AC gateway modules
Curtain control	Curtain motors, motor controllers, brackets
Sensors	Motion, temperature, humidity, door/window optional
Exterior	Outdoor relay panel, weatherproof enclosures
Garden	Outdoor light controller, irrigation provision
Safety	MCBs, surge protector, isolation, proper enclosures

22. Final Architecture Recommendation

Use a local-first smart home architecture with a central edge gateway, room-wise controllers, dedicated washroom controllers, an outdoor-rated exterior panel, and secure cloud access only for remote operation.

The design should avoid dependency on the internet for basic home operation. Lighting, fans, curtains, AC control, washroom exhausts, and exterior scenes should all continue working locally through wall switches and the local gateway.